

ZHIJING WU

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GitHub: <https://github.com/Zhijing-W>

EDUCATION

Columbia University

M.S. in Computer Science

New York, NY

Aug. 2025–Dec. 2026

University of Birmingham

*B.Sc. in Applied Mathematics with Mathematics | **First Class Honours***

Birmingham, UK

Sept. 2021–Jun. 2025

Jinan University

*B.Sc. in Mathematics and Applied Mathematics | **GPA: 3.84/4.25***

Guangzhou, CN

Sept. 2021–Jun. 2025

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL, JavaScript, Ruby, HTML/CSS, C, Bash/Shell, R, Matlab

Infrastructure & Data: GCP, Docker, Kubernetes, Kafka, PostgreSQL, MySQL, Redis, ClickHouse, Spark, TCP/IP, BGP, IS-IS, Routing Protocols, OSI Model

Framework & Tools: Spring Boot, Ruby on Rails, Microservices, REST APIs, Git, CI/CD, UNIX/Linux, OOD

WORK EXPERIENCES

iFLYTEK Co., Ltd

Hefei, CN

System Maintenance and Data Intern

Jul. 2023–Sept. 2023

- Engineered an **ETL** pipeline using **Python/Pandas** to standardize **12,000+** records of City Supplementary Health Insurance data. Automated format conversions and cleaning logic, **increasing data integrity to 98%**.
- Optimized **ClickHouse** analytical performance by designing the **OLAP schema** and deploying critical Materialized Views. This architecture pre-aggregated key financial metrics (market share, premium trends).
- Wrote analytical **SQL** (Window Functions, Array Functions) to compute core competitor intelligence metrics. These insights directly supported the strategic positioning of **2** new product iterations.

PROJECTS

Real-Time Job Search Tracker (Full-Stack) | [Github](#)

Aug. 2025–Present

- Architected a **full-stack** job application tracking platform using **Ruby on Rails (MVC)** framework, managing **100+** applications across **7 pipeline stages** with real-time updates.
- Implemented comprehensive testing strategy with **RSpec** (unit) and **Cucumber** (integration), achieving **95% code coverage** (SimpleCov) across **100+** test cases validating core user workflows.
- Engineered a canonical data anchoring system that consolidated application URLs from any job platforms (like LinkedIn, Handshake, Indeed, etc.) into a single source of truth, **eliminating 3-5 minutes of manual data entry** per application and auto-generating interactive Sankey diagrams to visualize application flow.

Community Trading Microservices System | [Github](#)

Sept. 2025–Present

- Designed **distributed** trading platform on **Google Cloud Platform (GCP)** with **5 microservices** exposing **30+ RESTful endpoints**, deployed across **Cloud Run (serverless)** and **Compute Engine (VM)** managing 4 independent databases (Cloud SQL + VM MySQL) with logical foreign key constraints.
- Integrated **OAuth 2.0 + JWT authentication**, **API Gateway**, **RESTful API** patterns (eTag, HATEOAS, asynchronous processing), and browser UI on **Cloud Storage** accessing distributed services.
- Engineered **idempotent** transaction endpoints with request deduplication to prevent duplicate charges, ensuring **data integrity** and consistency in distributed architecture.

RESEARCH EXPERIENCES

An Electric Vehicle Sales Hybrid Forecasting Method

Hefei, CN

Research Assistant, Anhui University, Prof. Jinpei Liu

Jan. 2022–Oct. 2024

- Built automated data collection pipelines with **BeautifulSoup** and **Requests** to scrape EV sales data and **10,000+ customer reviews** across **20** EV companies.
- Developed a sentiment analysis model combining **BERT** and **bidirectional LSTM**, improving accuracy and domain adaptability via training on diverse datasets.
- Achieved **R² of 0.9971**, surpassing XGBoost by **15.71%**, non-decomposed models by **11.31%**, and other sentiment methods by **1.92%**.